

AI Implementation Learning Roadmap

Goal: Learn how to build real AI-powered software by implementing progressively more advanced projects. Focus on understanding every line of code instead of copying tutorials.

Project 1 - Console AI Chatbot

- Build a terminal chatbot using the OpenAI API.
- Learn: API keys, .env, SDK installation, requests, responses, conversation history, error handling.
- Deliverable: Console chatbot with multi-turn conversation.

Project 2 - AI Text Analyzer

- Analyze job descriptions, emails, reviews and products.
- Return validated JSON using structured outputs.
- Learn: schemas, Pydantic, backend integration.

Project 3 - AI Appointment Assistant

- Use function/tool calling.
- Implement `get_available_slots()`, `create_appointment()`, `cancel_appointment()`.
- Learn: authorization, tool execution, AI + real application data.

Project 4 - Semantic Search

- Create embeddings for your own notes.
- Implement cosine similarity and vector storage.
- Learn how semantic search works before RAG.

Project 5 - PDF RAG Assistant

- Upload PDFs, extract text, chunk documents, generate embeddings, retrieve relevant chunks and answer with citations.
- Learn the complete RAG pipeline.

Project 6 - Agentic RAG

- Add an AI agent that decides when retrieval is needed.
- Implement query rewriting, routing, retrieval evaluation and citations.

Project 7 - Production AI Application

- React frontend + FastAPI/.NET backend.
- Authentication, PostgreSQL, streaming responses, logging, token usage, rate limiting, Docker, tests and deployment.

Learning Rule: Spend about 20% of your time on videos/documentation and 80% building, debugging and improving your own projects.